

Investment Applications of NLP: Does 10-K / 10-Q Sentiment Predict Stock Returns?

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Overview

This project tests whether the **sentiment expressed in corporate SEC filings (10-K and 10-Q)** carries information about *future* stock returns. The hypothesis: qualitative, firm-specific tone is harder for the market to price quickly than hard quantitative metrics, so a measurable lag may exist between when sentiment is disclosed and when it is fully reflected in price — a lag a systematic strategy could potentially capture.

To test this, we built and backtested **three long-short sentiment strategies of increasing complexity** over a 30-year window (1994–2023, ~242,600 firm-month observations) and compared their out-of-sample performance.

Headline finding: the *simplest* signal was by far the strongest. Added complexity — more sentiment categories, sector-neutralization, winsorization, and weight caps — consistently *hurt* performance rather than helping it.

Data

Component	Description
Filings	SEC 10-K and 10-Q disclosures
Sentiment lexicon	Loughran-McDonald financial dictionary — 7 categories: Negative, Positive, Uncertainty, Litigious, Strong Modal, Weak Modal, Constraining
Returns	Monthly equity returns, merged to filings via CIK / PERMNO
Window	January 1994 - December 2023 (360 months)
Sample	~242,600 firm-month observations

Filing word counts were used to normalize raw sentiment counts so that longer disclosures did not mechanically dominate the signal.

Methodology

Each strategy forms a monthly long-short portfolio and holds it for the following month's return.

1. Baseline strategy

A deliberately simple signal: the spread between positive and negative word counts, scaled by uncertainty.

Baseline Sentiment = ((N_Positive - N_Negative) / N_Uncertainty) + 1

Portfolio: long firms with a positive score (> 0), short firms with a negative score.

2. Expanded LM Quintiles

A richer model using all seven Loughran-McDonald categories. Scores were normalized by filing length and standardized into monthly **z-scores** so firms could be compared fairly within each month, preventing any single firm from dominating on naturally larger raw values.

Portfolio: sort firms into quintiles each month; go long the highest-sentiment quintile (F5) and short the lowest (F1).

3. Enhanced Sector-Neutral Capped

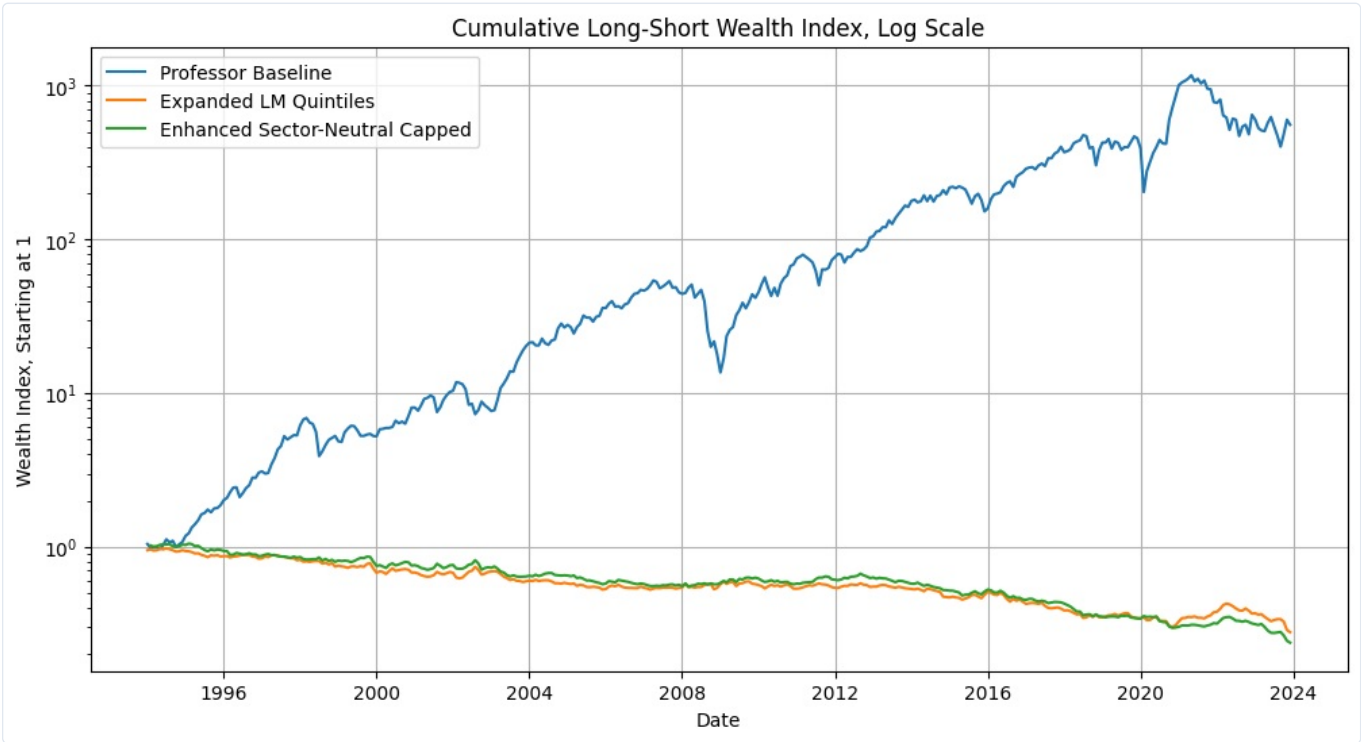
A refinement of the expanded model intended to strip out noise and concentration risk:

- **Sector-neutral ranking** using Fama-French industry classifications (FFInd) to remove industry-language bias.

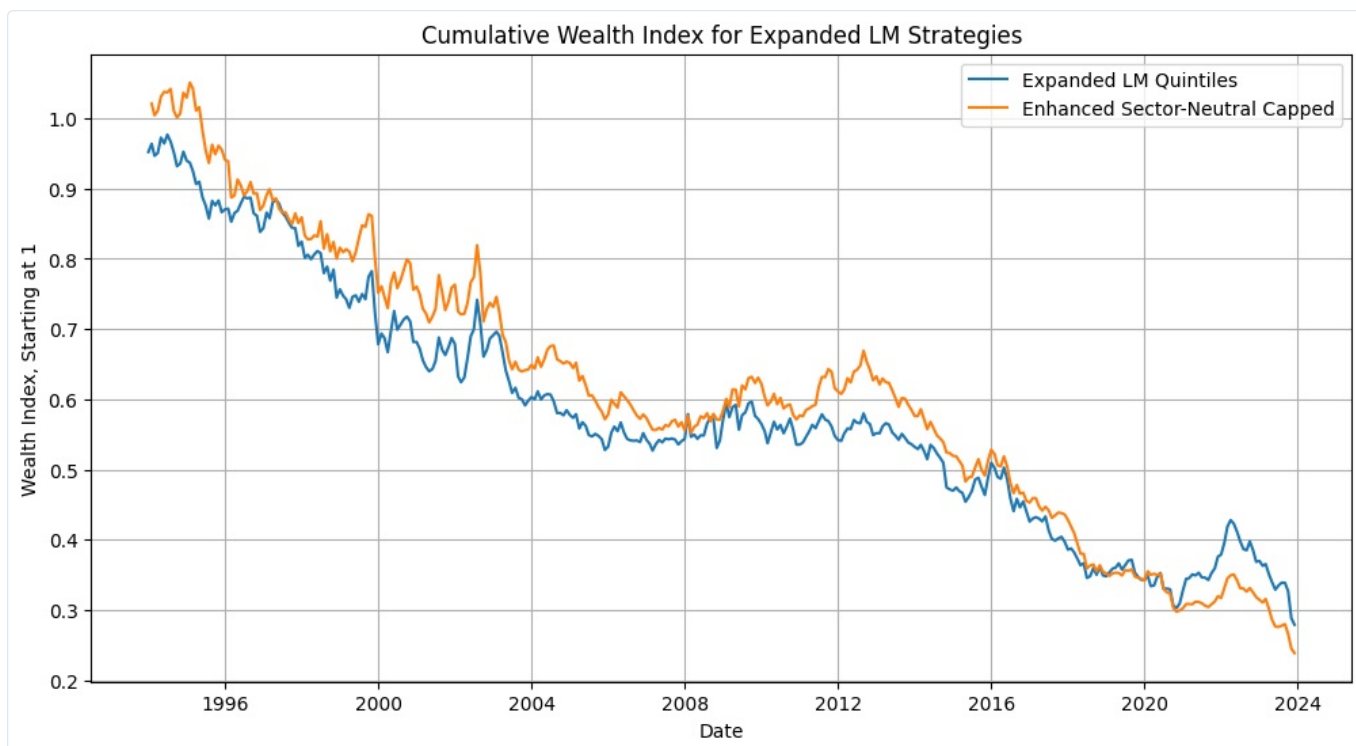
- **Winsorization** of the top and bottom 1% of sentiment scores to limit outlier influence.
- **Signal-strength weighting**, capped at 5% of the portfolio per name, to reward conviction without over-concentrating.

Results

Metric	Baseline	Expanded LM Quintiles	Enhanced Sector-Neutral Capped
Avg. monthly long-short return	+2.31%	−0.32%	−0.37%
Annualized return	+23.46%	−4.17%	−4.68%
Hit rate	65.56%	44.72%	42.90%
Sharpe ratio	0.78	−0.45	−0.58
Information ratio	0.23	−0.13	−0.17
Max drawdown	−74.81%	−71.46%	−77.32%
Average IC	−0.0058	−0.0081	−0.0090
Cumulative return	+55,628%	−72.12%	−76.16%

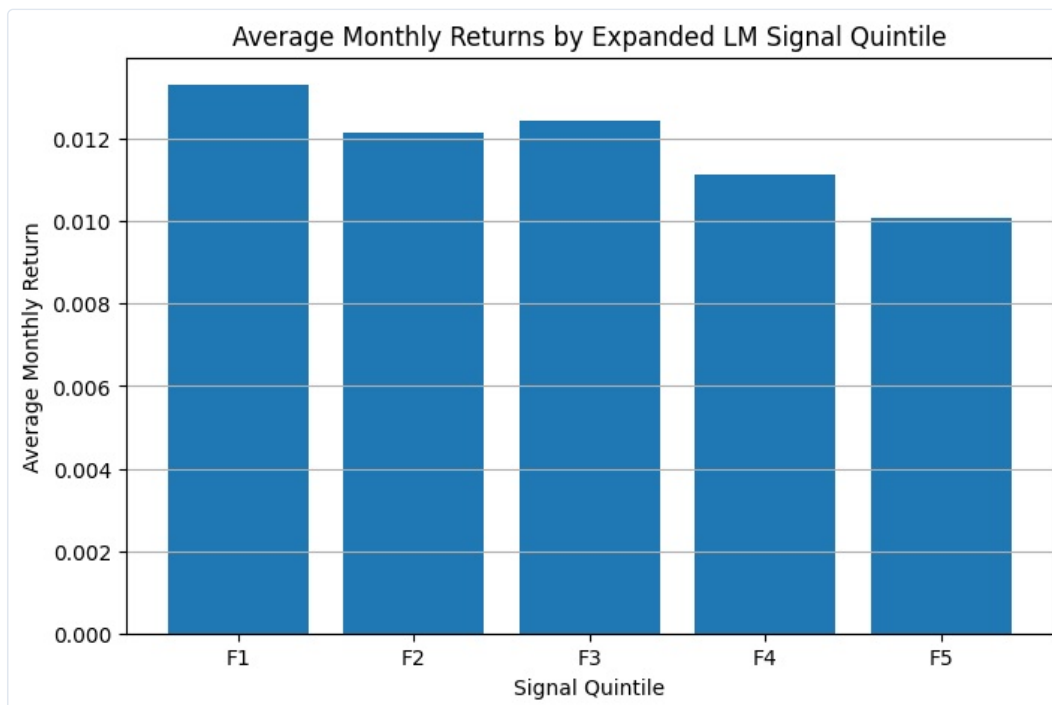


Cumulative long-short wealth index (log scale). The baseline compounds to roughly 500× initial wealth while both complex models decay below their starting value.

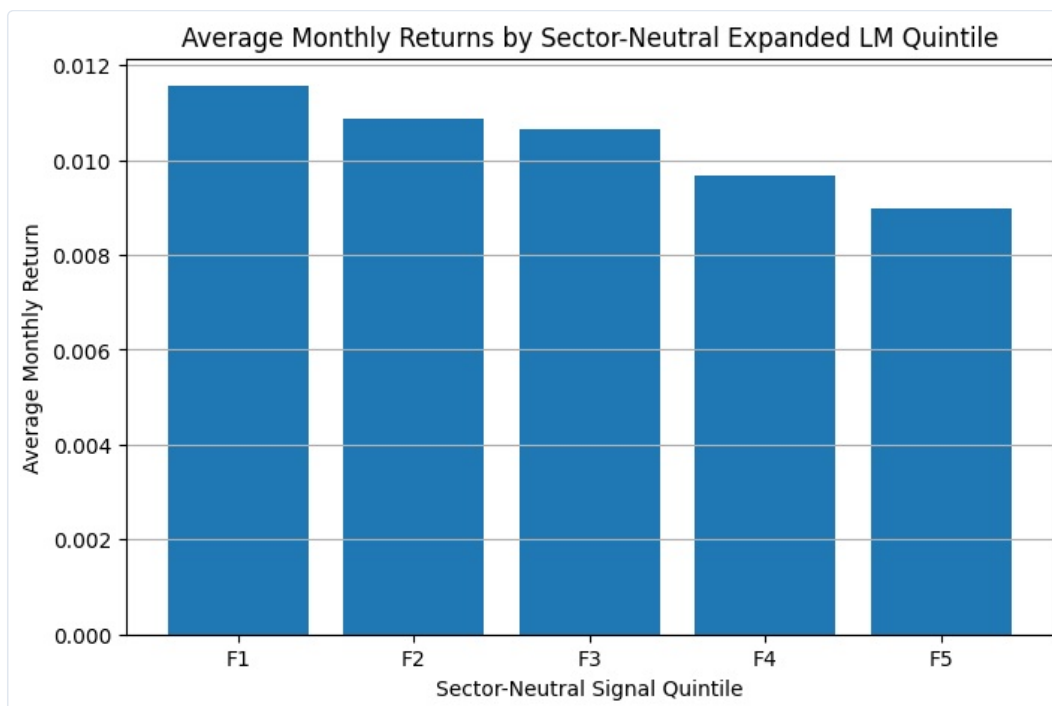


Isolating the two complex models shows their persistent, grinding decline.

Tellingly, the expanded and enhanced models show returns *declining* from the low-sentiment quintile (F1) to the high-sentiment quintile (F5) — the opposite of the intended long-high / short-low design, which is why their long-short spreads came out negative.



Average monthly returns by Expanded LM signal quintile.



Average monthly returns by Enhanced signal quintile.

Panel regression

Regressing next-month returns on each sentiment signal across all ~242,600 observations:

Model	Coefficient	t-stat	p-value	Adj. R ²	Observations
Baseline	−0.00141	−2.58	0.0099	0.110	242,599
Expanded LM	−0.00052	−4.63	0.000004	0.110	242,663

Both signals are statistically significant but carry small, *negative* coefficients at the firm-month level — consistent with the near-zero information coefficients and with the idea that low-sentiment firms are sometimes underpriced rather than simply "bad."

Discussion

Why did the simple baseline win? The baseline directly captures the positive-minus-negative tone spread and held up across the full sample, especially when annualized, supporting the core idea that disclosure tone has genuine economic relevance.

Why did complexity hurt? All three strategies posted near-zero (slightly negative) information coefficients. For the complex models, the more negative IC suggests they over-captured negative sentiment — which can signal *underpricing* rather than poor fundamentals. Low sentiment may flag riskier firms or pessimistic pricing (a risk premium / buying opportunity) rather than future underperformance, so aggressively shorting the lowest-sentiment names worked against the portfolios.

Drawdowns. Every strategy suffered a drawdown exceeding 70%. The baseline's worst stretch came after the 2008 crisis; the complex models' sharpest declines clustered around 2000 and 2003. Signals underperformed most during downturns, when sentiment is noisiest. Firm size, volatility, and portfolio construction all amplified drawdowns: if shorted names rally while sentiment fails to reflect it, losses compound quickly.

Limitations & Future Work

- A near-75% maximum drawdown is likely intolerable for a real investor; a sentiment signal would need to be combined with other factors and explicit risk controls.
- A **contrarian model** targeting the pessimism / underpricing premium — paired with a risk-only benchmark — could turn

the observed negative relationship into a usable edge.

- Swapping the Loughran-McDonald lexicon for a **transformer model such as FinBERT** may capture context and tone more accurately than dictionary counts.
- Adding **volatility controls, momentum, and disclosure-surprise features** could plausibly produce a genuinely profitable model.

Conclusion

A sentiment-based investing strategy is viable, but only as one input among several rather than a standalone signal. The most important lesson was methodological: **a more complex model is not always a better one**. The simplest baseline captured the real, persistent value in disclosure tone, while each layer of added sophistication introduced noise that degraded performance.

Tech stack: Python, pandas, NumPy, statsmodels, matplotlib, Loughran-McDonald sentiment dictionary. | Full pipeline and backtest available in the accompanying Jupyter notebook.